

C Tasks

Week 3

Q1: 1-Can a structure contain different data types? Give an example.

2-What is a nested structure in C? Provide an example

3-What happens if you assign one structure variable to another of the same type?

4-What is the purpose of the typedef keyword in structures? Give an example.

5-What is the difference between structure and union in C? Give an example of every one of them.

6-How do you use a pointer to a structure? Provide an example.

7- What is padding and packing in structures? How can you prevent structure padding in C?

Q2: 1-implement a structure that stores student name and his marks in the different three subjects (Arabic, English, math) as integers and take number of students and data of students as input. Hint (size of structure must be as small as possible and student name will not exceed 19 characters).

2- write function to sort them based on the sum of degrees of these subjects and if two students have the same sum of degrees sort them based on their name. Hint (you may use bubble sort algorithm).

Output samples of Q2:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of students : 4
enter name of th1 student : ahmed
enter th1 student degrees in order (Arabic,English,math)10 20 30
enter name of th2 student : hassan
enter th2 student degrees in order (Arabic,English,math)20 40 100
enter name of th3 student : ali
enter th3 student degrees in order (Arabic,English,math)10 20 30
enter name of th4 student : samy
enter th4 student degrees in order (Arabic,English,math)70 80 0
the students sorted in increasing order based on sum of degrees :
ahmed
ali
samy
hassan

Process returned 0 (0x0)    execution time : 26.932 s
Press any key to continue.
```

Q3: 1-Implement a structure that represents a complex number which has two integer parts imaginary and real and function to print this complex number.

2-write functions to do addition and subtraction and multiplication of complex numbers using functions. Hint (you will use four functions to evaluate this expression).

And calculate $(2+5i)^7 + (3+7i) - (4+6i)$.

Q4:

- 1- Define an Enum for traffic light colors (RED, YELLOW, GREEN) and write function to print a corresponding action using a switch statement. Hint(handle if Enum variable takes a value out of the range).
- 2- Define a struct Employee with Enum Role to store employees' job positions {engineer, manager, hr } and string to store his name and print the details. What is required is to take number of employees as input and there data then print them.
- 3- What does #pragma do in C? , What is the role of #ifdef and #ifndef? , What does #pragma once do? How is it different from include guards?
- 4- Write #define for pi that is used to calculate circumference, area, and volume of spheres.

Math equations : <https://www.1728.org/diamform.htm>

Q5:

1- Write a program to store an integer and a character in a union and print both.

2- Explain how unions can be used for efficient memory usage in embedded systems.

3- Implement a union-based polymorphic function that takes a single union parameter and determines whether it contains an integer, float, or string based on a flag. Hint(you may use an enum that has {int,float,str} to pass to the function with union and may use switch case).